

Experiment - 2

Data Wrangling II

Create an "Academic performance" dataset of students and perform the following operations using Python.

1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.
2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.
3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution. Reason and document your approach properly.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [2]: import pandas as pd
```

```
In [3] df = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\2\AcademicPerformance.csv")
```

In [4]:

```
print(df)
```

```
gender race/ethnicity parental level of education lunch \
0 female group B bachelor's degree standard
1 female group C some college standard
2 female group B master's degree standard
3 male group A associate's degree free/reduced
4 male group C some college standard
...
2235 NaN NaN NaN NaN NaN
2236 NaN NaN NaN NaN NaN
2237 NaN NaN NaN NaN NaN
2238 NaN NaN NaN NaN NaN
2239 NaN NaN NaN NaN NaN

test preparation course Year_Birth math score reading score \
0 none 1970.0 72.0 72
1 completed 1961.0 NaN na
2 none 1958.0 90.0 95
3 none 1967.0 NaN NaN
4 none 1989.0 76.0 78
...
2235 NaN NaN NaN NaN
2236 NaN NaN NaN NaN
2237 NaN NaN NaN NaN
2238 NaN NaN NaN NaN
2239 NaN NaN NaN NaN

writing score Dt_Admission College_Fees
0 74 6/16/14 $84,835.00
1 A 6/15/14 $57,091.00
2 93 5/13/14 $67,267.00
3 44 05-11-2014 $32,474.00
4 75 04-08-2014 $21,474.00
...
2235 NaN NaN NaN
2236 NaN NaN NaN
2237 NaN NaN NaN
2238 NaN NaN NaN
2239 NaN NaN NaN
```

[2240 rows x 11 columns]

In [5]:

```
print(df['math score'])
```

```
0    72.0
1     NaN
2    90.0
3     NaN
4    76.0
...
2235 NaN
2236 NaN
2237 NaN
2238 NaN
2239 NaN
Name: math score, Length: 2240, dtype: float64
```

```
In [6]: print(df['math score'].isnull())
```

```
0      False
1       True
2      False
3       True
4      False
...
2235    True
2236    True
2237    True
2238    True
2239    True
Name: math score, Length: 2240, dtype: bool
```

```
In [7]: print(df['reading score'])
```

```
0      72
1     na
2     95
3    NaN
4     78
...
2235  NaN
2236  NaN
2237  NaN
2238  NaN
2239  NaN
Name: reading score, Length: 2240, dtype: object
```

```
In [8]: print(df['reading score'].isnull())
```

```
0      False
1      False
2      False
3       True
4      False
...
2235    True
2236    True
2237    True
2238    True
2239    True
Name: reading score, Length: 2240, dtype: bool
```

```
In [9]: missing_values = ["n/a", "na", "--"]
df = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\2\AcademicPerformance.csv", na_v
```

```
In [10]: print(df['reading score'])
```

```
0      72.0
1       NaN
2      95.0
3       NaN
4      78.0
...
2235    NaN
2236    NaN
2237    NaN
2238    NaN
2239    NaN
Name: reading score, Length: 2240, dtype: float64
```

```
In [11]: print(df['reading score'].isnull())
```

```
0      False
1       True
2      False
3       True
4      False
...
2235     True
2236     True
2237     True
2238     True
2239     True
Name: reading score, Length: 2240, dtype: bool
```

```
In [12]: dataset = [11,41,20,3,101,55,68,97,99,6]
```

```
In [13]: sorted(dataset)
```

```
Out[13]: [3, 6, 11, 20, 41, 55, 68, 97, 99, 101]
```

```
In [14]: quantile1, quantile3 = np.percentile(dataset, [25,75])
```

```
In [15]: print(quantile1, quantile3)
```

```
13.25 89.75
```

```
In [16]: iqr_value = (quantile3 - quantile1)
```

```
In [17]: print(iqr_value)
```

```
76.5
```

```
In [18]: lower_bound_value = quantile1 - (1.5*iqr_value)
```

```
In [19]: upper_bound_value = quantile3 + (1.5*iqr_value)
```

```
In [20]: print(lower_bound_value, upper_bound_value)
```

```
-101.5 204.5
```

```
In [21]: from datetime import date
df['age'] = date.today().year - df['Year_Birth']
```

```
In [22]: df['Year'] = pd.DatetimeIndex(df['Dt_Admission']).year
df['E_L'] = date.today().year - df['Year']
```

```
In [23]: df.head(5)
```

Out[23]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	Year_Birth	math score	reading score	writing score	Dt_Admi
0	female	group B	bachelor's degree	standard	none	1970.0	72.0	72.0	74	6
1	female	group C	some college	standard	completed	1961.0	NaN	NaN	A	6
2	female	group B	master's degree	standard	none	1958.0	90.0	95.0	93	5
3	male	group A	associate's degree	free/reduced	none	1967.0	NaN	NaN	44	05-11
4	male	group C	some college	standard	none	1989.0	76.0	78.0	75	04-08

```
In [24]: df['Fees$'] = df['College_Fees'].str.replace(',', '').str.replace('$', '').str.replace(
df['Fees_M$'] = df['Fees$'].apply(lambda X:round(X/1000000))
```

```
In [25]: df.head(5)
```

Out[25]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	Year_Birth	math score	reading score	writing score	Dt_Admi
0	female	group B	bachelor's degree	standard	none	1970.0	72.0	72.0	74	6
1	female	group C	some college	standard	completed	1961.0	NaN	NaN	A	6
2	female	group B	master's degree	standard	none	1958.0	90.0	95.0	93	5
3	male	group A	associate's degree	free/reduced	none	1967.0	NaN	NaN	44	05-11
4	male	group C	some college	standard	none	1989.0	76.0	78.0	75	04-08

In []: